

PART 1

Q1 What is the start date and end date available in each dataset?

	Dataset	first date	last date
0	Brent Oil	1946-01-01	2026-03-01
1	Gold	1915-01-01	2026-04-01
2	Silver	1915-01-01	2026-04-01
3	swift_reports	2025-12-01	2026-03-01

Q2 How many rows are available for Brent Oil, Gold, Silver and the Swift Currency Tracker data?

	Dataset	rows count
0	Brent Oil	963
1	Gold	1336
2	Silver	1336
3	swift_reports	288

Q3 Are there any missing values or repeated dates? Summarise what you

	Dataset	Total Missing Values	Repeated Dates
0	Silver	0	0
1	Gold	0	0
2	Brent Oil	0	0

found.

Q4 Convert the Date column into a proper date format and create Month and Year columns for the commodity datasets.

```
def convert_date(df):  
    df["Date"] = pd.to_datetime(df["Date"])  
    df["Month"] = df["Date"].dt.month  
    df["Year"] = df["Date"].dt.year  
    return df  
  
silver_df = convert_date(silver_df)  
gold_df = convert_date(gold_df)  
oil_df = convert_date(oil_df)
```

With this code date column was converted.

Q5 For the Swift dataset, identify the main metrics available, such as global payment share, trade finance share or offshore RMB economy share.

Both Global Payment Share and offshore Share are important. And may be used separately for different analysis.

Q6 What were the highest and lowest Brent Oil prices in the dataset, and in which years did they occur?

	lowest_v	L_date	heighest_v	H_date
0	21.11	1946-01-01	211.21	2026-03-01

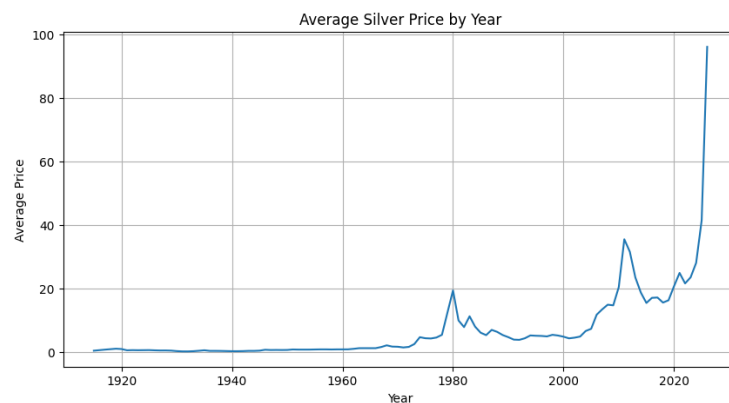
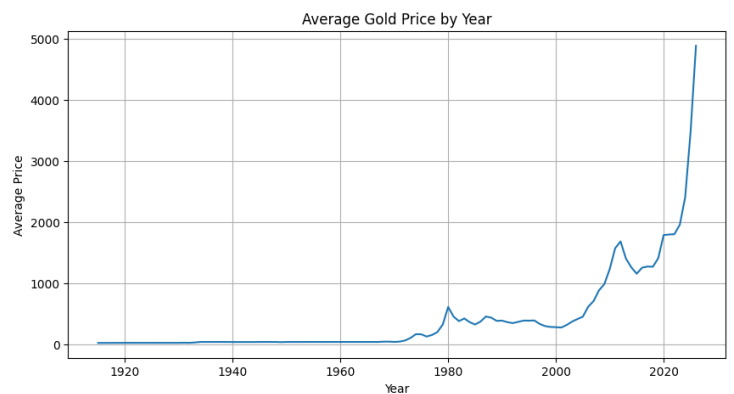
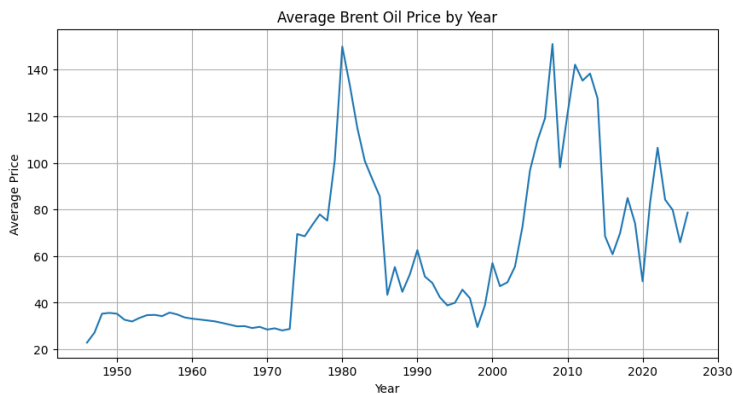
Q7 What were the highest and lowest Gold prices in the dataset, and in which years did they occur?

	lowest_v	L_date	heighest_v	H_date
0	19.25	1915-01-01	5277.89	2026-04-01

Q8 What were the highest and lowest Silver prices in the dataset, and in which years did they occur?

	lowest_v	L_date	heighest_v	H_date
0	0.28	1915-01-01	113.9528	2026-04-01

Q9 Calculate the average price by year for Brent Oil, Gold and Silver.



Avg year prices for dataset

Q10 Which five years had the highest average Brent Oil price?

YEAR	AVERAGE_BRENT_OIL_PRICE
2008	150.96
1980	149.86
2011	142.12
2013	138.30
2012	135.27

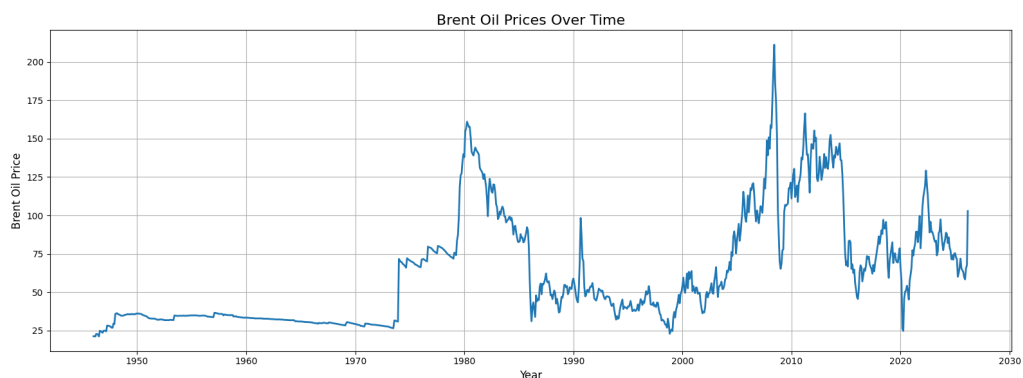
Q11 Which five years had the highest average Gold price?

date_date	avg_value
2026	4881.619800
2025	3472.540975
2024	2404.577567
2023	1953.688517
2022	1798.956575

Q12 Which five years had the highest average Silver price?

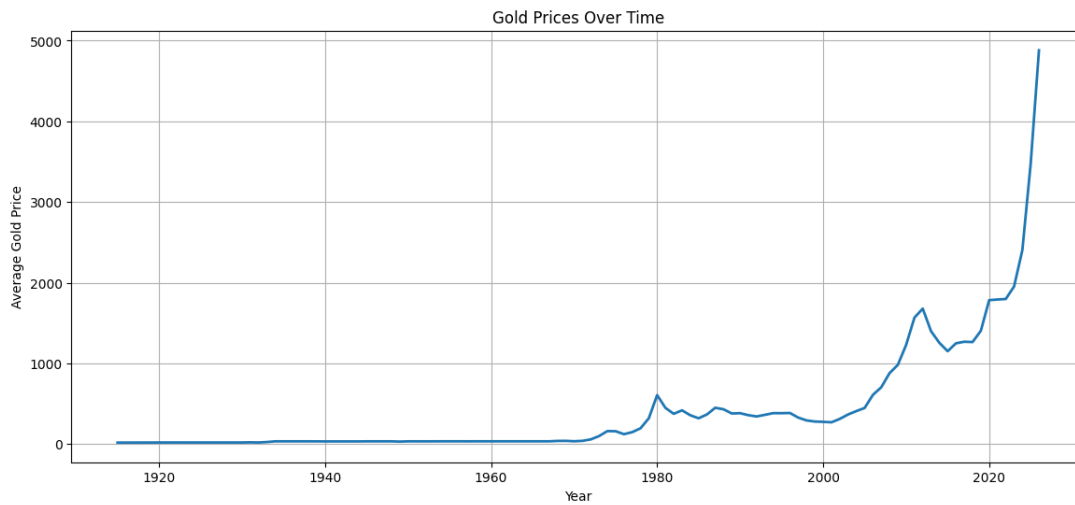
date_date	avg_value
2026	96.083200
2025	41.501500
2011	35.556667
2012	31.626667
2024	28.128033

Q13 Create a line chart for Brent Oil prices over time and describe the main periods of increase and decline.



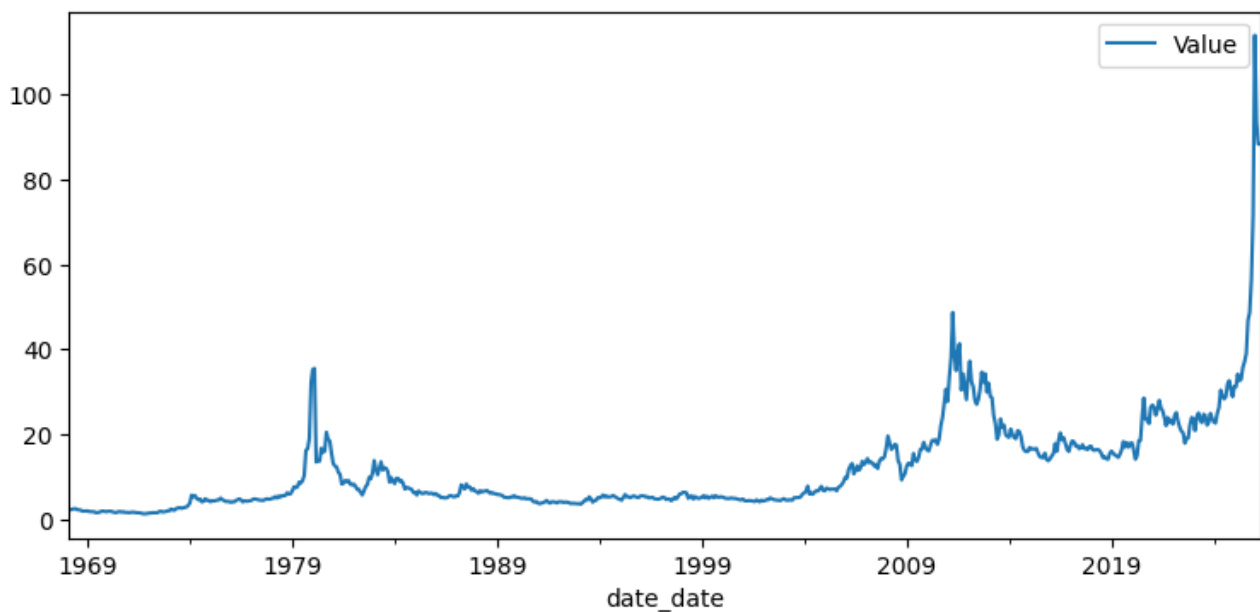
1946-1973 – “Oil prices remained relatively stable in early years.”,
1973-1980 – “Oil crisis and supply shocks increased prices.”,
1981-1998 – “Market adjustment and lower demand reduced prices.”,
1999-2008 – “Global demand and economic growth increased prices.”,
2008-2009 – “Global financial crisis reduced demand.”,
2010-2014 – “Prices recovered after the financial crisis.”,
2014-2016 – “Oil oversupply caused a major price fall.”,
2020 – “COVID-19 reduced global oil demand.”,
2021-2022 – “Economic recovery and supply issues increased prices.”

Q14 Create a line chart for Gold prices over time and describe the long term trend.



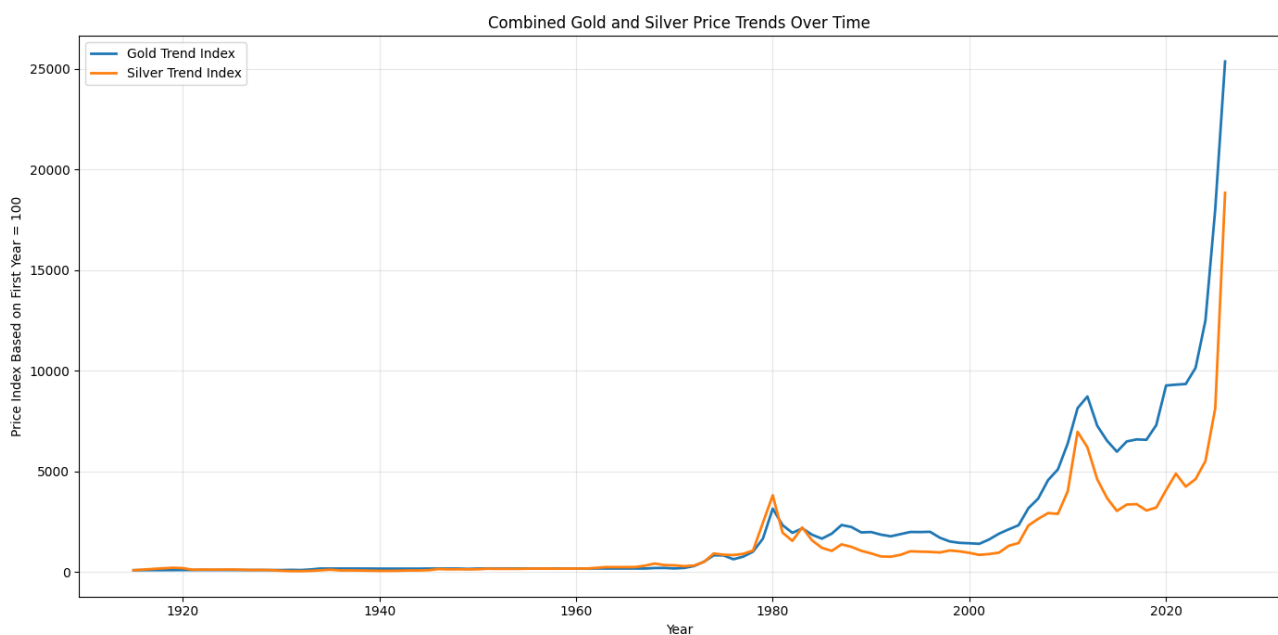
Crisis causes gold prices to spike. Overall long-term trend is increasing gold prices.

Q15 Create a line chart for Silver prices over time and describe the main peaks.



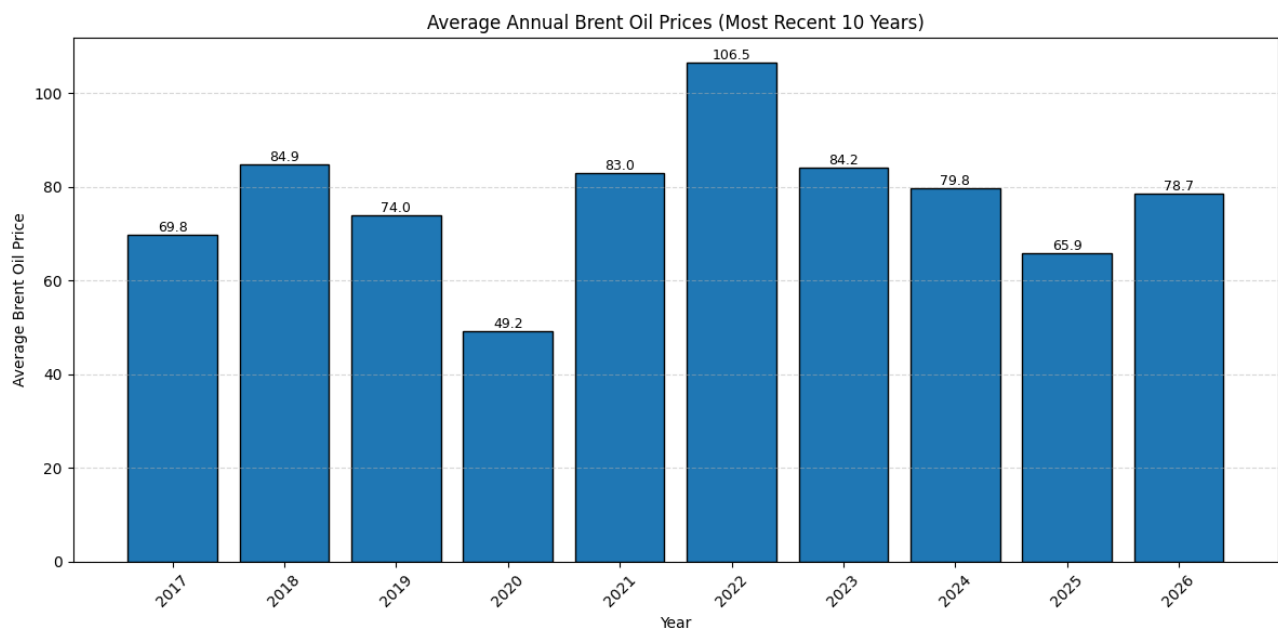
- 1) 1980 spike was made by Hunter Brothers, they got 70% of world silver (<https://www.britannica.com/topic/Silver-Thursday>) prices dropped March 27, 1980
- 2) demand was high after crisis 2008. Buying of metals was becoming easier year to year. It crashed 2011
- 3) Very small spike after 2019 is covid
- 4) spike after 2024 is war (Iran) and overall anxiety in global markets

Q16 Create one combined chart showing Gold and Silver trends. What similarities or differences can you observe?



Both gold and silver react to world crisis in a similar way.

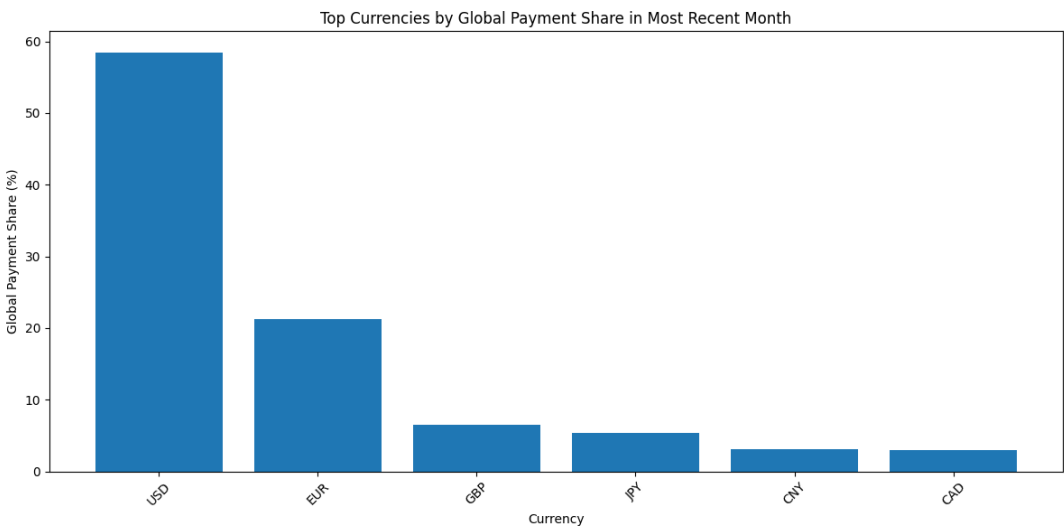
Q17 Create a bar chart showing average annual Brent Oil prices for the most recent 10 years in the dataset



Q18 Create a simple table showing the latest available value for Brent Oil, Gold and Silver.

Commodity	Latest_Date	Latest_Value
Brent Oil	2026-01-03	102.86
Gold	2026-01-04	4712.89
Silver	2026-01-04	88.36

Q19 Using the Swift Currency Tracker file, identify the top currencies by global payment share in the most recent available month.



Q20 What is the USD share in the most recent available global payment data?
58.50%

Q21 What is the CNY/RMB share and rank in the most recent available global payment data?

Latest_Month	Currency	Global_Payment_Share	Global_Rank
2026-03	CNY	3.1	5.0
2026-03	CNY	2.44	6.0
2026-03	CNY		5.0
2026-03	CNY	8.04	
2026-03	CNY		4.0
2026-03	China	5.99	

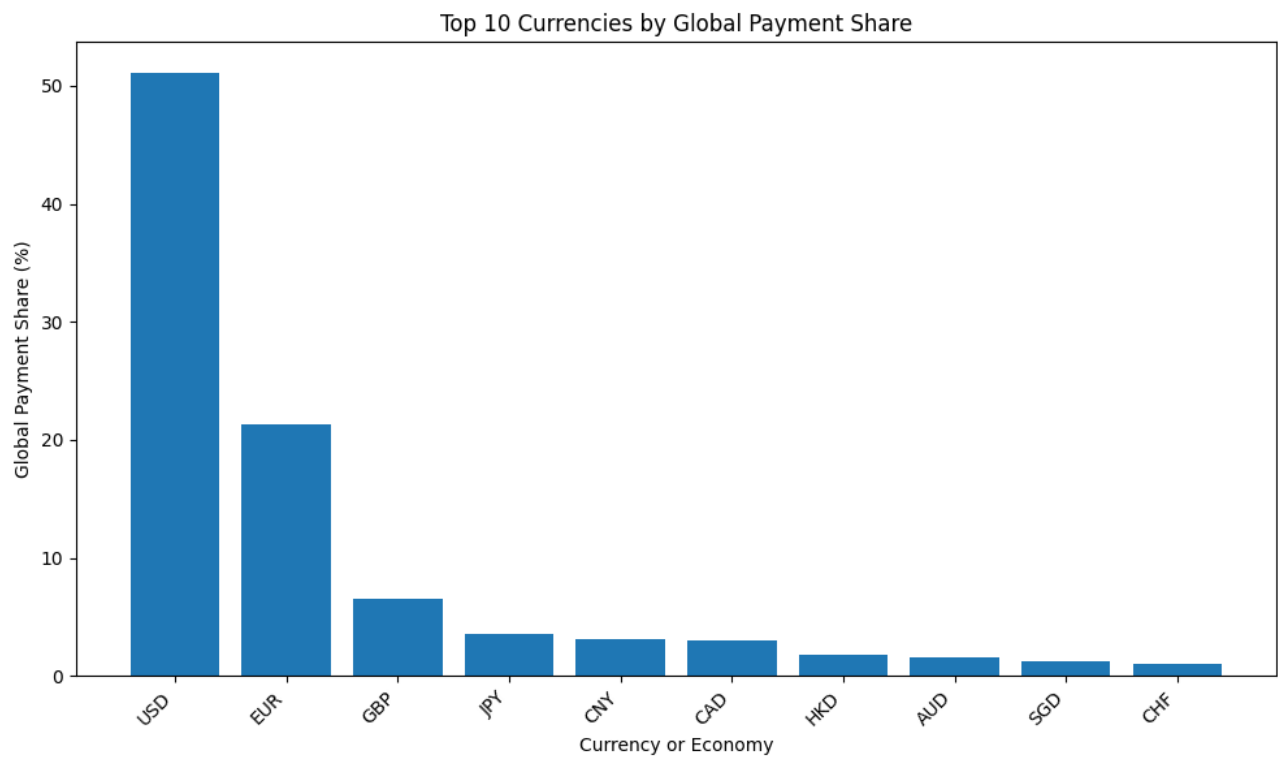
Q22 Which currencies appear most often in the top five across the available reports?

currency_or_economy Times_in_Top_5	
CNY	5
AUD	2
CAD	2
CHF	2
EUR	2
GBP	2
HKD	2
JPY	2
KRW	2
USD	2

Q23 Which offshore RMB economy has the highest share in the available offshore RMB data?

Offshore_RMB_Economy Highest_Share	
Hong Kong	75.57

Q24 Create a simple bar chart for the top 10 currencies or economies from one selected month.



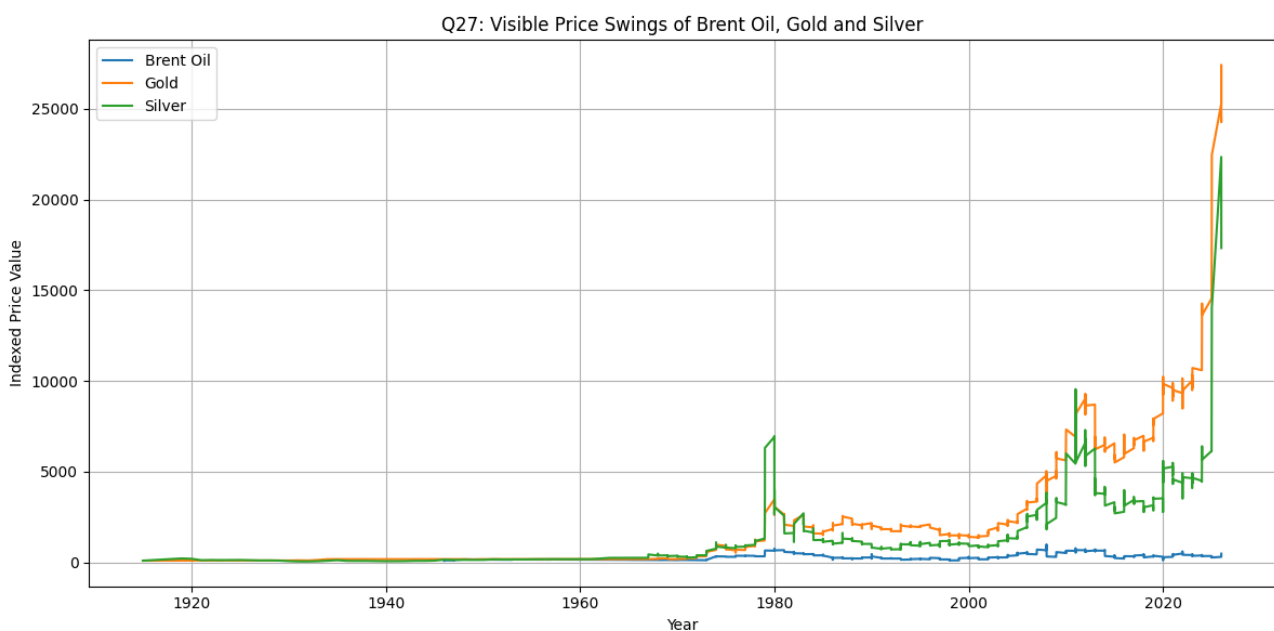
Q25 Compare the latest Gold value with its average value across the full dataset. Is it above or below the long term average?

Latest_Gold_Value		Long_Term_Average		Comparison
4712.89		397.55		Above Long-Term Average

Q26 Compare the latest Brent Oil value with its average value across the full dataset. Is it above or below the long term average?

Latest_Brent_Oil_Value		Long_Term_Average		Comparison
102.86		62.71		Above Long-Term Average

Q27 Which commodity appears to have the largest visible price swings: Brent Oil, Gold or Silver? Support your answer with one chart or summary table.

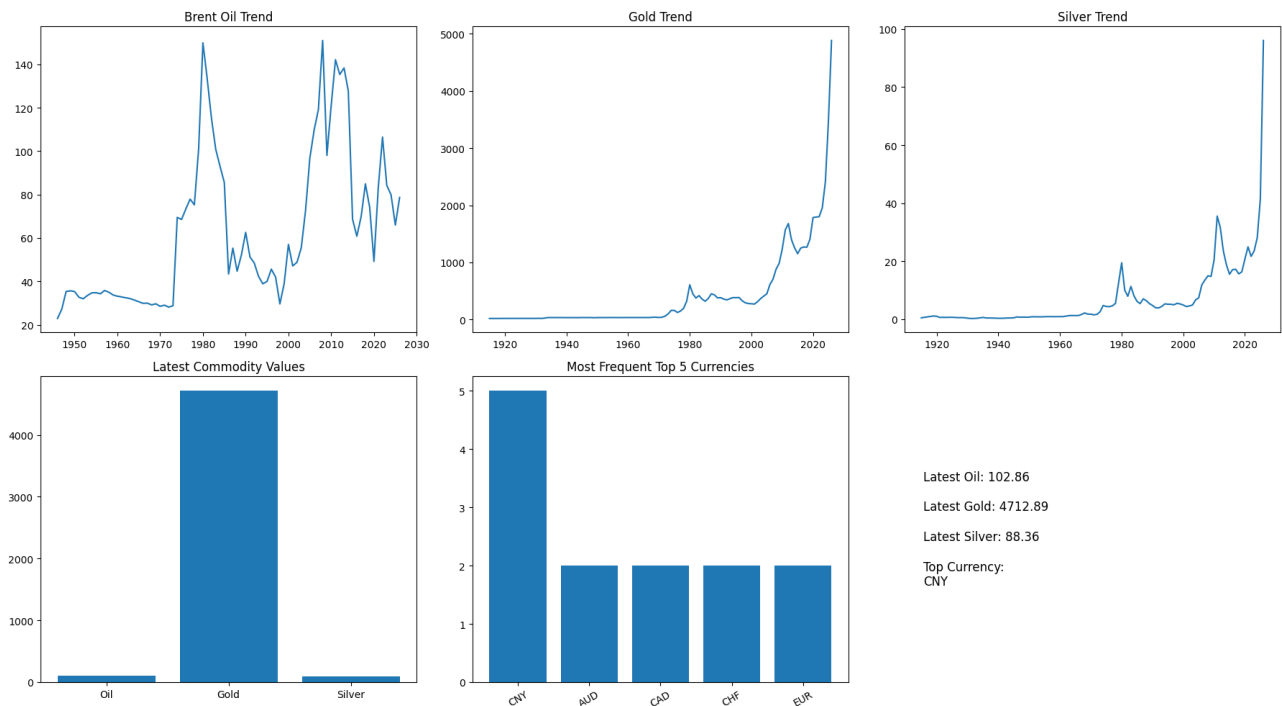


Silver has bigger swings, than gold, than oil

Q28 Choose one year with a strong oil price movement and check whether gold or silver also changed noticeably in that year.

Year	Oil_Change%		Gold_Change%		Silver_Change%
1974	141.55		61.75		83.37

Q29 Create a short dashboard page with 4 to 6 visuals summarising the most important findings.



Q30 Write three short insights that a manager could understand without technical knowledge.

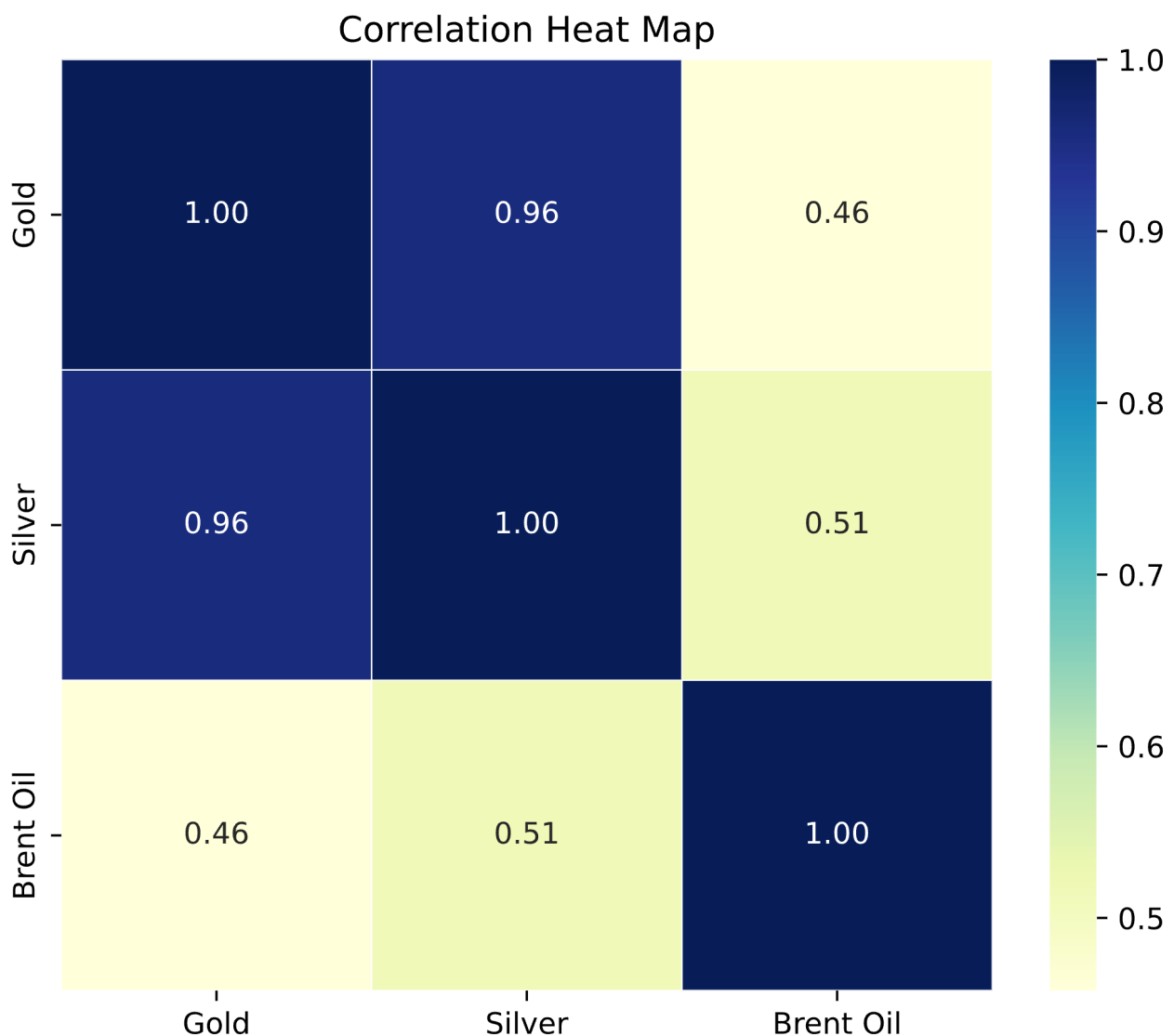
1. Brent Oil prices are continuously increase and after Iran-Isareal war, prices are to high in whole world.
2. Gold and Silver are always in demand therefore price of gold and silver increase day by day.
3. Few country currencies rule the whole market such as pounds, USD and euro.

BONUS QUESTIONS

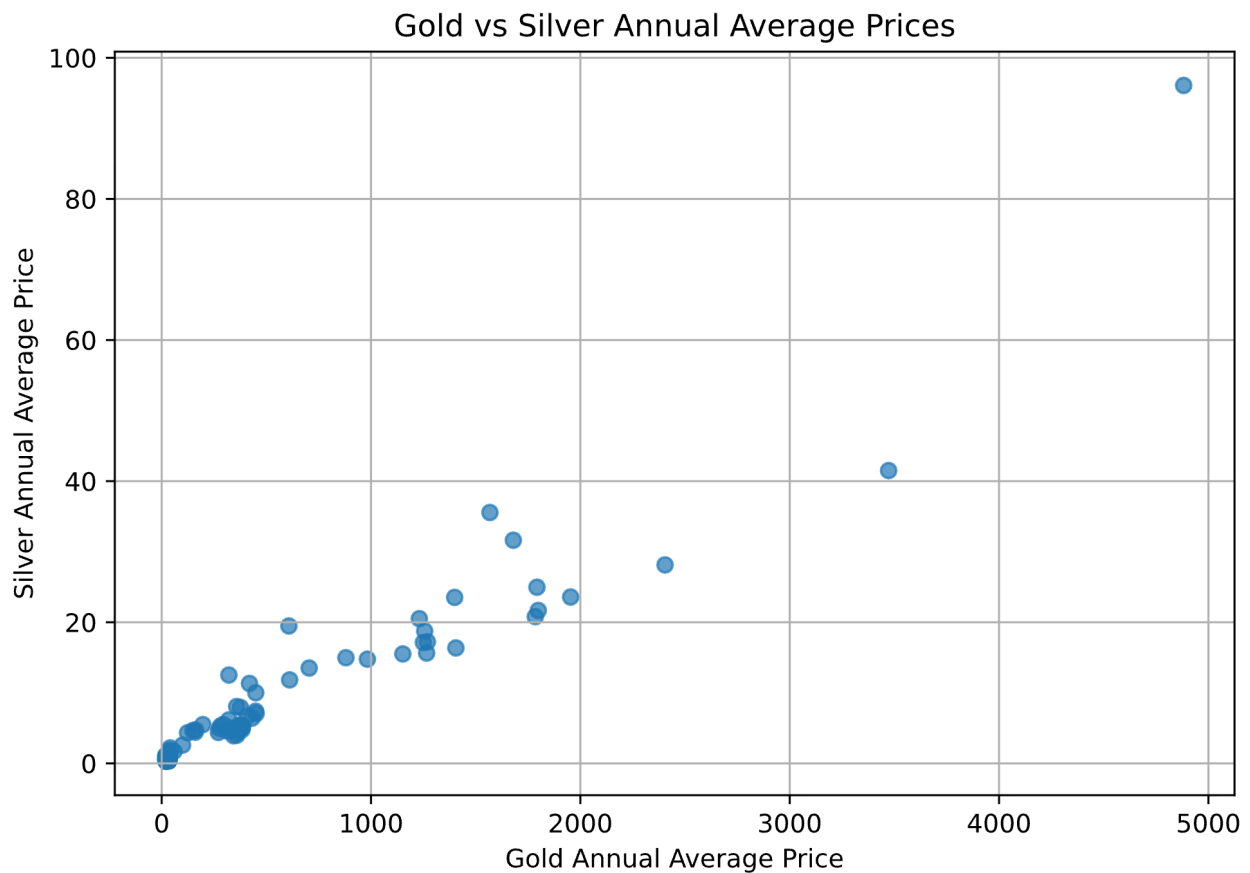
QB2 Create a simple correlation table between Gold, Silver and Brent Oil using annual average prices.

	Gold	Silver	Brent Oil
Gold	1.0	0.959	0.458
Silver	0.959	1.0	0.515
Brent Oil	0.458	0.515	1.0

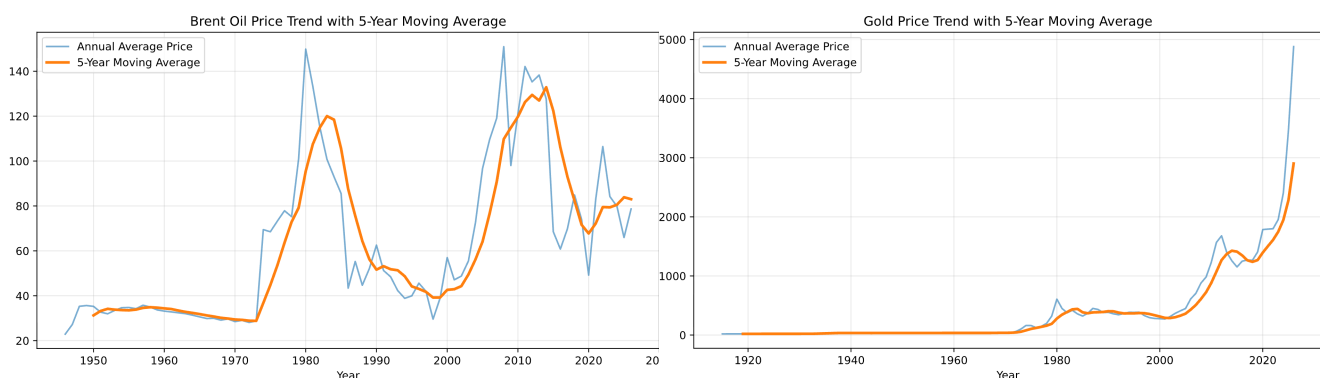
QB3 Create a heat map based on the correlation table.



QB4 Create a scatter plot comparing Gold and Silver annual average prices.

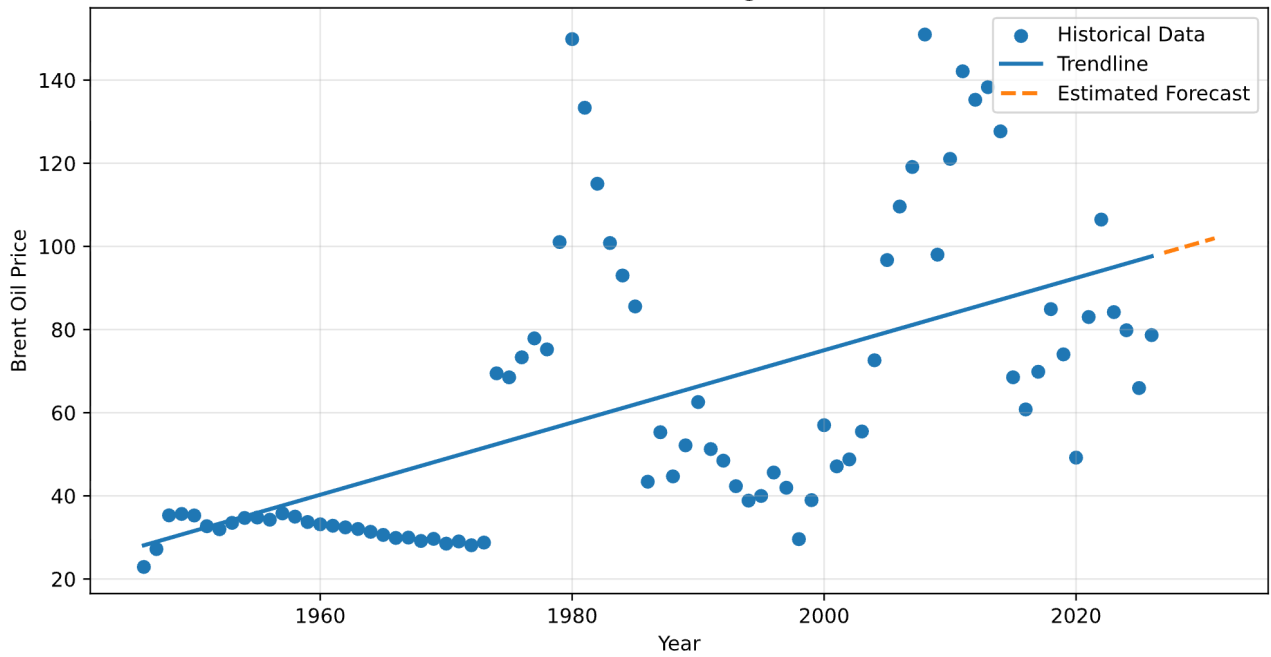


QB5 Use a moving average line to smooth the Gold or Brent Oil trend and explain what it shows.

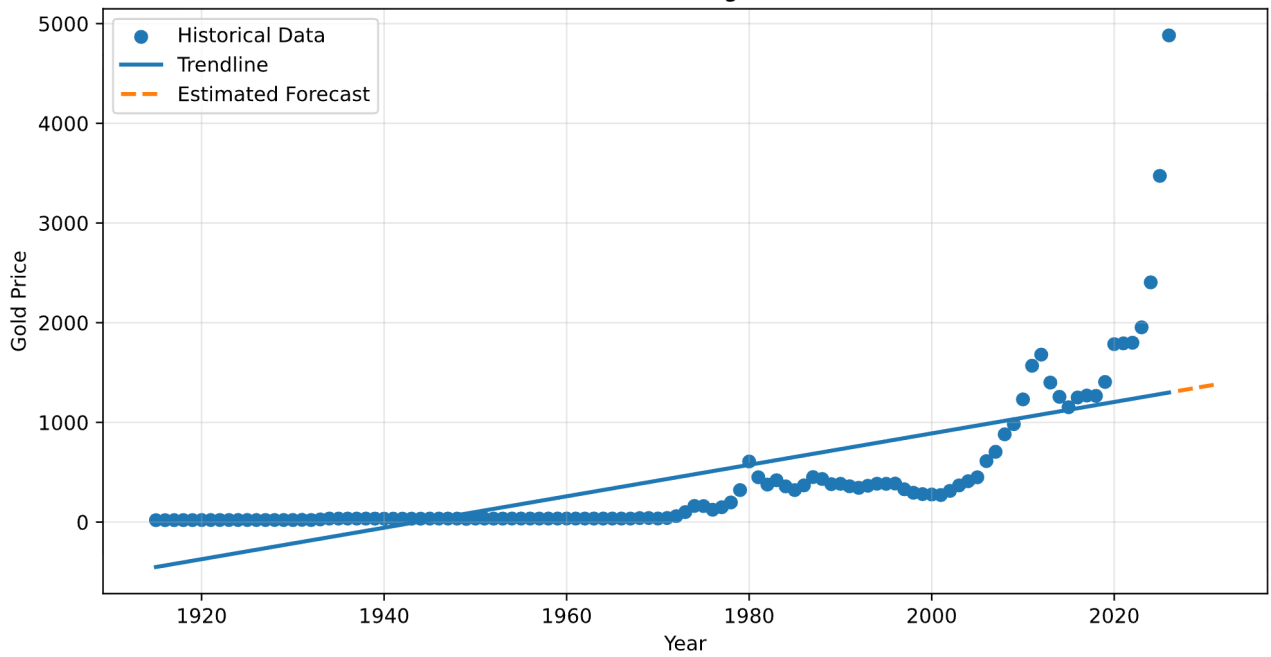


QB6 Create a simple forecast using a trendline and clearly state that it is only an estimate, not a guaranteed prediction.

Brent Oil Price Forecast Using Linear Trendline



Gold Price Forecast Using Linear Trendline



PART 2

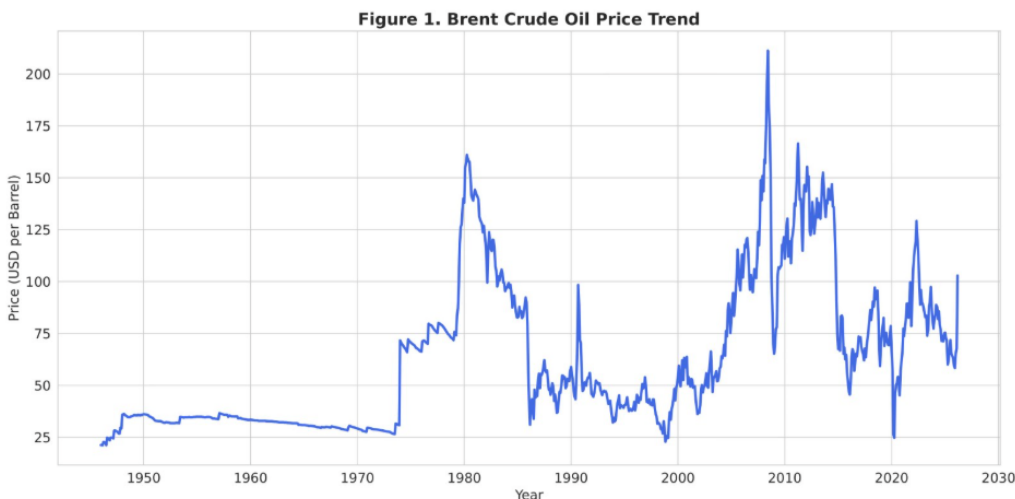
Q1 Identify three major global events mentioned in the files that could affect commodity or currency markets. (Amal)

The provided reports indicate three major events that could influence commodity and currency markets: (1) the Hormuz Strait crisis and energy supply disruptions, which may threaten global oil supplies; (2) sanctions and tariff discussions, which can affect international trade flows and currency demand; and (3) the emergence of BRICS Pay together with increased gold accumulation by central banks, which reflects efforts toward reserve diversification. The SWIFT Global Currency Tracker shows that although the U.S. dollar remains dominant, other currencies continue to participate in international payments and trade finance, indicating gradual changes in the global monetary system .

Q2 Create a simple timeline of selected events such as the Hormuz crisis, energy supply disruption, sanctions, tariff discussion, BRICS Pay, or gold accumulation. (Amal)

Event	Possible Market Impact
Hormuz Strait Crisis	Increased oil price volatility
Energy Supply Disruptions	Higher energy prices
Economic Sanctions	Changes in trade and currency movements
Tariff Discussions	Greater market uncertainty
BRICS Pay Development	Reduced reliance on traditional payment systems
Central Bank Gold Accumulation	Increased demand for gold and silver

Q3 Which event seems most directly connected to Brent Oil prices? Explain why. (Amal)



The Hormuz Strait crisis and related energy supply disruptions appear to be most directly connected to Brent Oil prices. Since a large share of global crude oil trade passes through this region, geopolitical tensions can create concerns about supply shortages and lead to higher oil prices. This relationship is reflected in **Figure 1 (Brent Crude Oil Price Trend)** from Part I, where periods of volatility correspond to supply-side uncertainties.

Q4 Which event seems most directly connected to Gold or Silver prices? Explain why. (Amal)



Central bank gold accumulation and the development of BRICS Pay appear to be most directly connected to Gold and Silver prices. These developments encourage reserve diversification and increase demand for precious metals, which are often considered safe-haven assets. This pattern is visible in **Figure 2 (Historical Silver Price Trend)** from Part I, where long-term price increases reflect rising demand and periods of economic uncertainty.

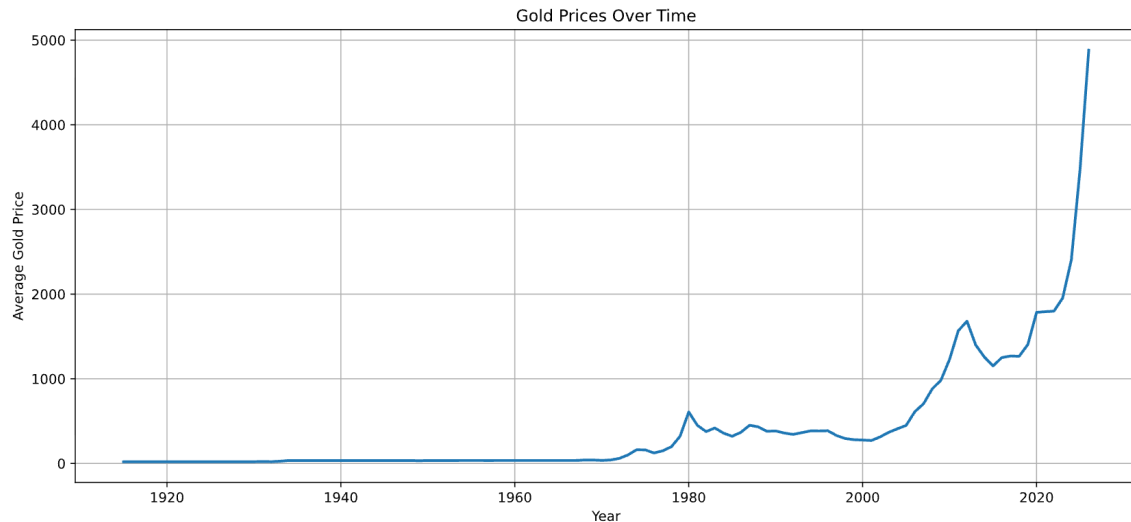
Q5 How could an oil supply shock or shipping disruption affect Brent Oil prices? Link your answer to one Brent Oil chart. (Anna)



In 1973, OPEC countries proclaimed an oil embargo. It created a huge demand, and therefore prices rose.
In 1980, the Iran/Iraq war decreased oil supplies again, which increased prices even more.

2014-2016: Oil oversupply caused a major price fall. OPEC has a mechanism to regulate oil prices by setting a maximum limit of oil production.

Q6 Why might Gold increase during geopolitical uncertainty? Link your answer to one Gold chart. (Anna)

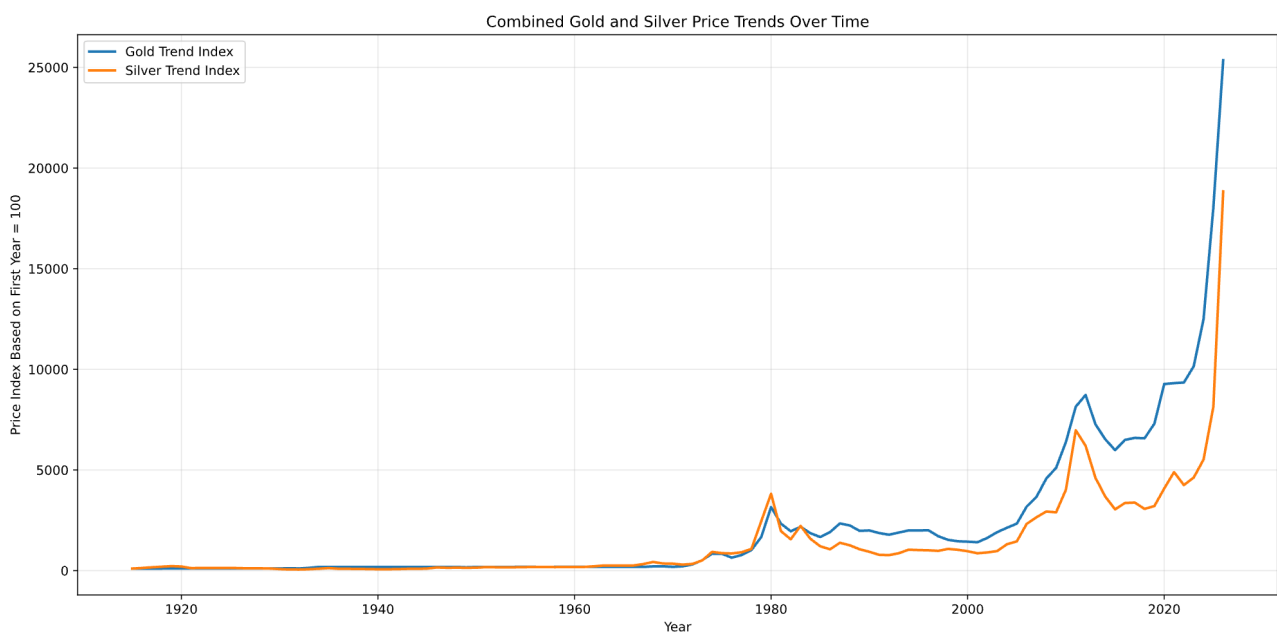


1980: Height inflation. Reason: investors' speculative demand.

2011: Several countries in the European Union were in default. That provoked interest in gold, which increased demand and therefore increased prices.

Gold considered to be a safe spot in the periods of uncertainty.

Q7 Is Silver behaving more like a precious metal or an industrial commodity in your selected period? Explain using your chart. (Anna)



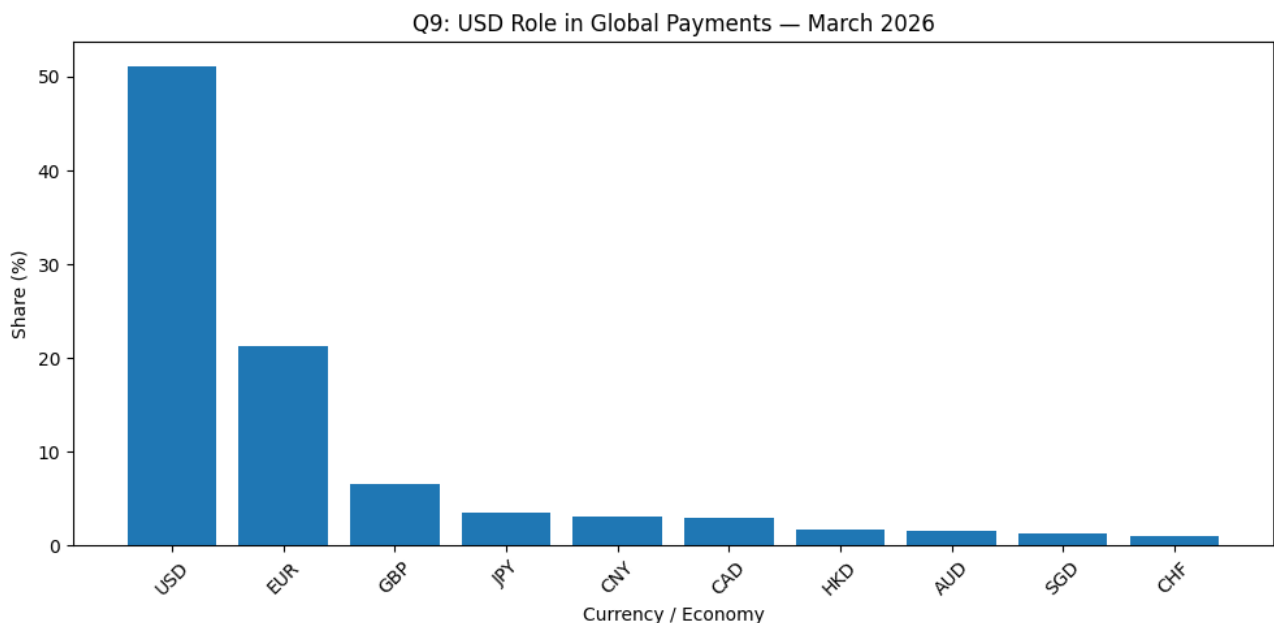
On a picture above, it is obvious that silver and gold has similar fluctuations. So for the selected period silver prices behaving like precious metal.

Q8 Choose one crisis period and compare the behaviour of Oil, Gold and Silver in that period. (Anna)

In 1980, the Iran/Iraq war decreased oil supplies again, which increased prices.
The war provokes height inflation.

During the crisis period prices for gold, silver and oil rose. After crisis there was a period of 5 years of decreasing of prices

Q9 How does the Global Currency Tracker show the role of USD in international payments?(Nithin)



Analysis month: March 2026 | Full dataset period: December 2025 to March 2026 | Source: Swift Global Currency Tracker

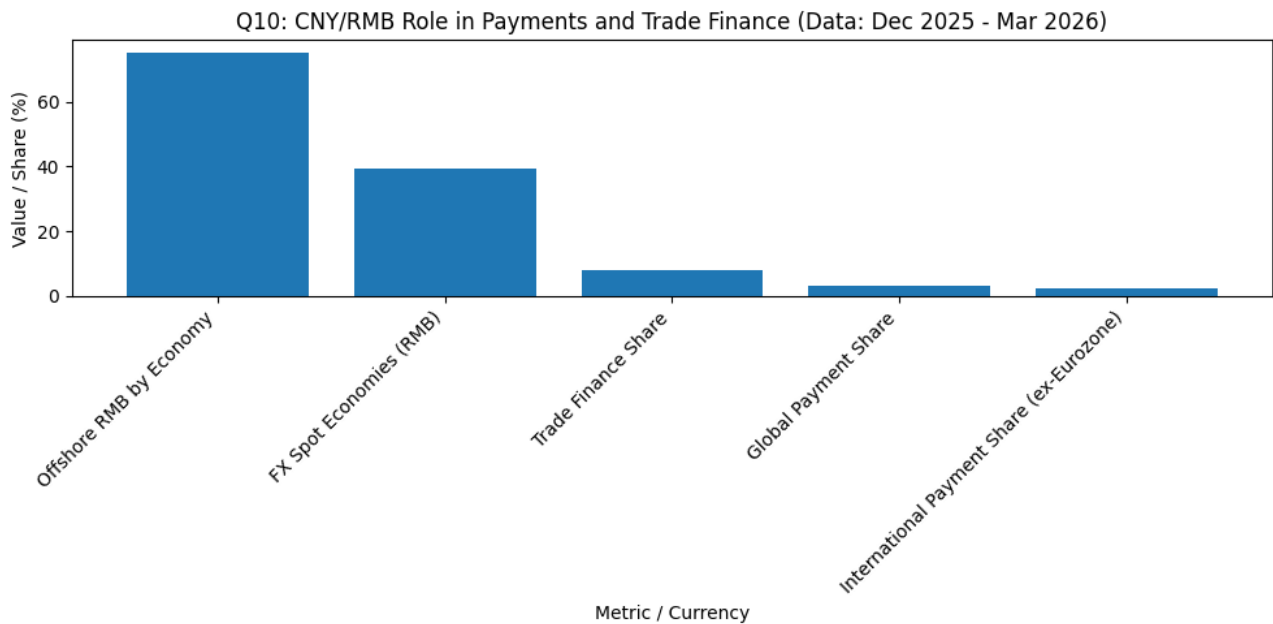
The analysis uses the latest available **data month: March 2026**.

The full CSV dataset available for this notebook covers **December 2025 to March 2026**.

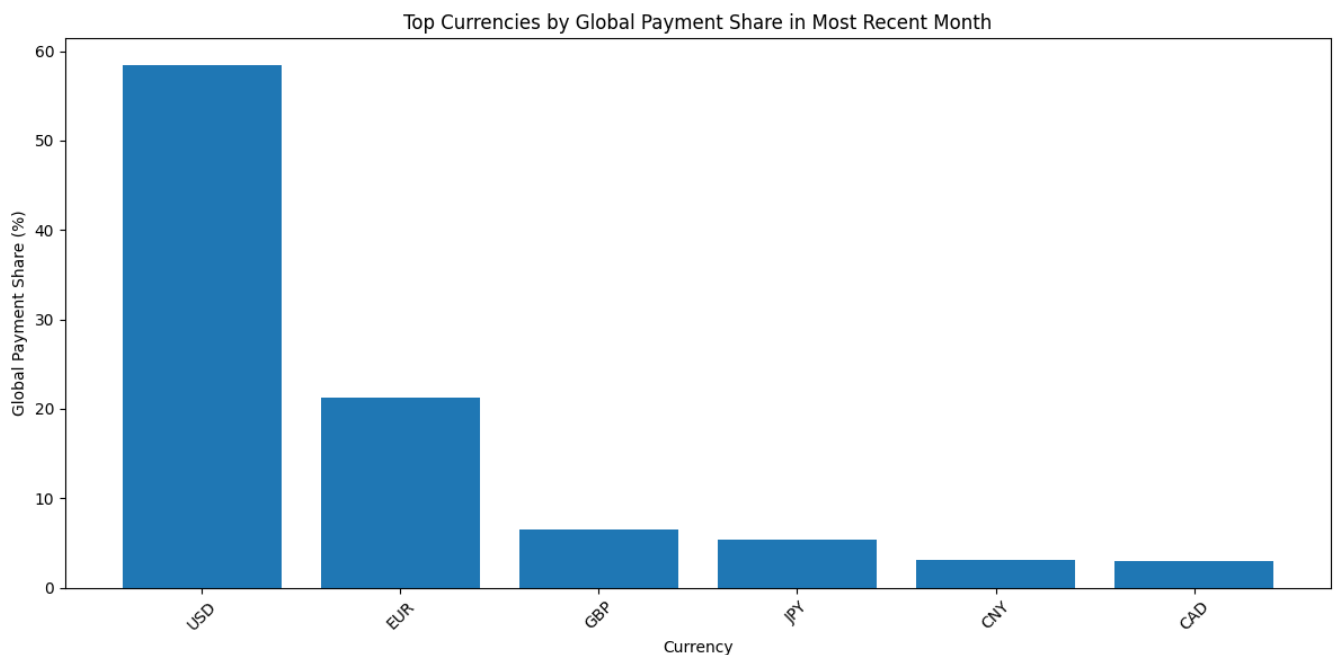
For March 2026, the Global Currency Tracker shows that USD has the highest global payment share among the listed currencies. Therefore, USD remains the dominant currency in global/international payments.

Q10 What does the tracker show about the role of CNY/RMB in global payments and trade finance?(Nithin)

The analysis uses Global Currency Tracker data from **December 2025 to March 2026**. The latest chart/table uses the **April 2026 report**, which reports **March 2026 data**. The tracker shows CNY/RMB's role through its share and ranking in global payments, international payments, trade finance, and FX spot activity. In the latest data, CNY/RMB appears in the tracker but remains much smaller than USD, showing that RMB is internationally used but not dominant.



Q11 Why might sanctions, alternative settlement systems, or BRICS Pay discussions matter for currency market analysis? (Nithin)



Sanctions, alternative settlement systems, and BRICS Pay discussions are important for currency market analysis because they influence how countries conduct international payments and settle trade. When countries face sanctions or seek to reduce reliance on USD-based payment systems, they may explore using local currencies, RMB/CNY, gold, or alternative payment networks. This shift can:

- Change the demand for major currencies
- Influence exchange rate behavior

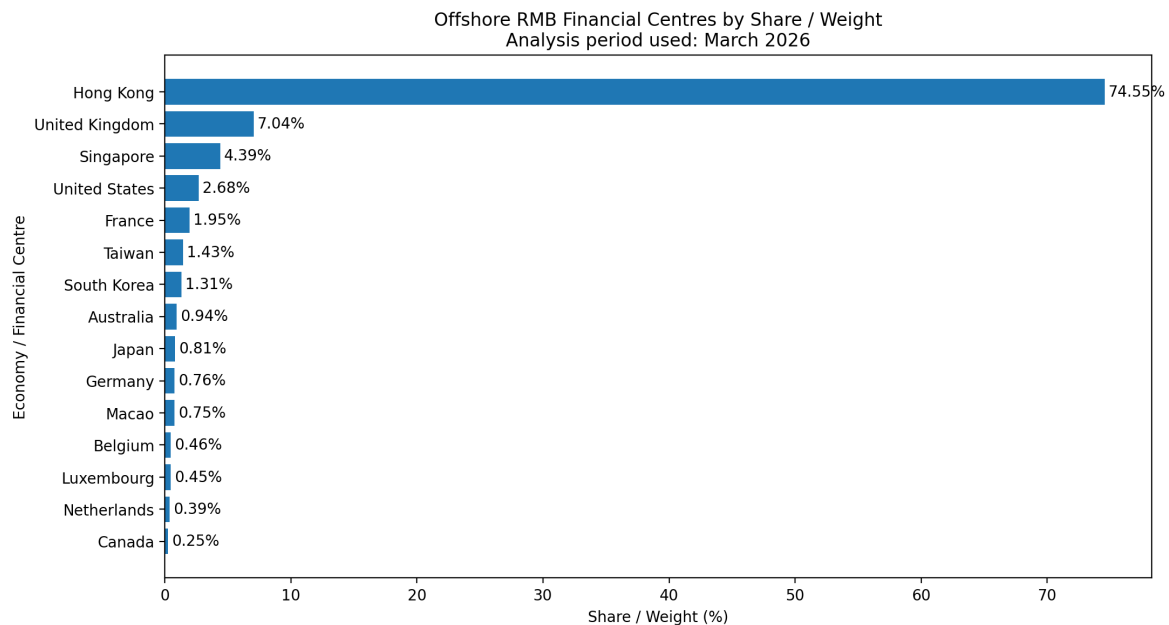
This topic connects to Part I Q19: Top currencies in global payment share, where the USD currently dominates, and the CNY has a much smaller share. Even minor moves towards systems

like BRICS Pay or other non-USD settlement methods could have significant implications for currency markets by:

Gradually reducing global USD dependence

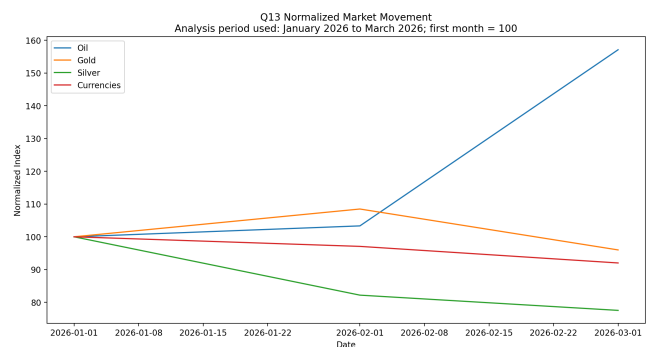
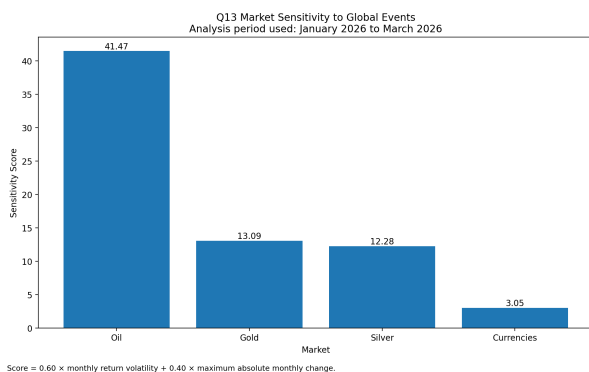
Increasing the role of alternative currencies in international trade

Q12 What does the offshore RMB data suggest about the role of Hong Kong and other financial centers? (Nithin)



The offshore RMB data for March 2026 suggests that Hong Kong is the main offshore hub for RMB payments. It ranks 1st with 74.55% of the offshore RMB share/weight, far above United Kingdom at 7.04% and Singapore at 4.39%. This shows that offshore RMB activity is highly concentrated in Hong Kong, while other financial centres play smaller supporting roles in the international use of RMB.

Q13 From your analysis, which market looks most sensitive to global events: Oil, Gold, Silver or Currencies? (Nithin)



Based on the analysis from January 2026 to March 2026, Oil looks like the most sensitive market to global events.

It ranks 1st with a sensitivity score of 41.47, monthly return volatility of 34.433%, and a maximum absolute monthly change of 52.025%.

This is higher than Gold, which ranks 2nd with a score of 13.086.

The result suggests that Oil reacted more strongly to global shocks during the common analysis period (January 2026 to March 2026), as shown in the sensitivity summary table and the market sensitivity score chart.

Q14 Write a short risk note for a company exposed to oil prices. (Nithin)

Companies that rely heavily on oil face substantial uncertainty because oil prices are highly volatile and can change rapidly in response to global events.

Factors such as supply chain disruptions, economic sanctions, wars, OPEC decisions, and shipping route issues all impact oil prices.

The Brent Oil month-year table (from January 1946 to March 2026) provides clear evidence of these dramatic price swings over time, emphasizing the ongoing risk. Business Impact:

Industries affected: Logistics, airlines, manufacturing, food delivery, and any company dependent on imports are especially vulnerable.

Consequences of price spikes: A sudden increase in oil prices can cause fuel, shipping, production, and supply costs to rise sharply, which erodes profit margins and can threaten business viability.

Risk Management Strategies:

Monitor Brent oil prices monthly to stay aware of trends and anticipate changes.

Use flexible pricing in contracts to adjust for cost increases and avoid locked-in losses.

Set aside emergency funds as a financial buffer for unexpected spikes in oil-related costs.

Diversify suppliers and shipping routes to reduce vulnerability to disruptions in any one region or route.

Consider hedging or locking in fixed fuel prices through financial contracts, especially when the global outlook is uncertain.

By proactively managing these risks, companies can better protect their profit margins and maintain operational stability even in volatile oil markets.

Q15 Write a short risk note for a company holding assets in USD, Gold or RMB/CNY. (Anna)

In today's world, China and the USA are two highly tangled economies and two trade partners. As it was discussed in the previous question, currencies are less volatile in the short-term perspective.

But in the long-term perspective, currencies are less reliable.

For China's economy, a strong local currency is deadly because it will make their products less competitive.

Gold is falling after a period of being on the rise; do not feel it would change in the short term.

Risks:

1 US President Trump.

- Trade wars with China can affect USD and CNY.

- A stronger RMB/CNY can reduce China's export competitiveness, so it has no potential to grow significantly.

2- USD currency as a reserve currency.

- For now, USD is still a reserve currency. But it must be monitored closely.

- After a new currency occurs, the reserve will become weaker.

3- Gold seems least risky, but grows with inflation.

- For the past 40 years, gold has been considered a long-term hedge against inflation.

- After the USA/Israel/Iran war, gold spiked and is now returning to previous prices.
- Price can be volatile in the short and medium term.

Summary:

Short-term USD is the best option.

Long-term Gold is the best option.

RMB/CNY is not a good option for now, but may be an option for diversification

Q16 Give three practical recommendations for investors, companies, or policymakers based on your findings. (Anna)

Recommendation for investors:

Diversify your investments.

Investors should maintain diversified portfolios across different asset classes and geographic regions. A combination of equities, gold, and other assets can help reduce exposure to currency fluctuations and geopolitical risks. Companies with global operations may provide additional resilience during periods of economic uncertainty.

- I would recommend individual investors to invest in gold and US companies with access to global market

Recommendation for companies:

Companies should diversify both their markets and supply chains. Expanding operations across multiple regions can reduce dependence on a single economy, currency, or trading partner.

Geographic diversification may improve resilience against trade disputes, regulatory changes, and local economic downturns.

Recommendation for policymakers:

Policymakers should focus on maintaining economic stability, predictable regulations, and sustainable long-term growth. Investments in infrastructure, innovation, and productivity can strengthen economic competitiveness and increase investor confidence. Consistent and transparent policies help build trust among domestic and international stakeholders.